Popularising Summary

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Master's thesis: an extremal problem on connectivity in networks (Erdős Problem 915).

Many of the networks we rely on are useful precisely because they offer backup routes. If one road is closed, traffic can take another. If one cable fails, a message can still reach its destination along a different chain of links. Mathematicians study this kind of structure with graphs, where dots stand for objects and lines stand for the connections between them. This thesis asks a simple-sounding question about such networks: how many connections can a network with a fixed number of points have, if we forbid any two points from being joined by too many separate, non-overlapping routes?

That question is a version of a problem posed decades ago by the mathematicians Béla Bollobás and Paul Erdős, listed today as Erdős Problem 915. In its original form it concerns ordinary networks and routes that are not allowed to share a connection. For that case the answer has been known since a theorem of Mader. This thesis starts from that result and then changes the rules of the network one at a time: connections may be doubled up, directions may be added so that a link works only one way, or a single connection may join more than two points at once. With each change the pencil-and-paper proof becomes longer and more delicate, until checking it by hand stops being realistic.

At that point the thesis hands the work to a computer. It builds a single program that can describe every version of the network in the same way, measure exactly how many separate routes run between any two points, and prove the answer outright for small networks by checking every possibility. For the larger networks, where checking everything is hopeless, it instead searches for good examples, either by imitating the way hot metal cools into an orderly shape, drifting at first and then locking into place, or by a stricter method that always takes the best step available and keeps a short memory of where it has just been so that it cannot walk in circles. Reassuringly, when either search is pointed at the cases we already understand, it rediscovers the very networks the earlier proofs had singled out.

The interplay between proof and experiment then pays off, in four places. For networks whose connections may each tie several points together at once, the thesis determines the exact largest possible network under the two strictest limits on backup routes, whatever the total number of points and however many points each connection joins. For one-way networks it proves, with no assumption about what the best network looks like, that the largest possible number of links is a quarter of the square of the number of points, give or take an amount growing only in proportion to that number. That settles the shape of the answer where even the leading term had been a conjecture, and what is still unknown there is a single numerical factor of about eight. And for one-way networks whose links may be doubled up, it determines the exact largest possible network completely, whatever the total number of points and whatever the limit on backup routes. The key idea is to find, inside any such network, a thin skeleton that alone remembers who can reach whom, throwing every doubled-up link away but one. That skeleton is never large, and removing it knocks every pair's backup-route count down by exactly one, so peeling off skeleton after skeleton, one for each unit of the limit, accounts for every link and proves the count outright. A fourth case follows: for one-way networks with many-point connections, the obvious two-layer design is, for large networks, the best there is.